



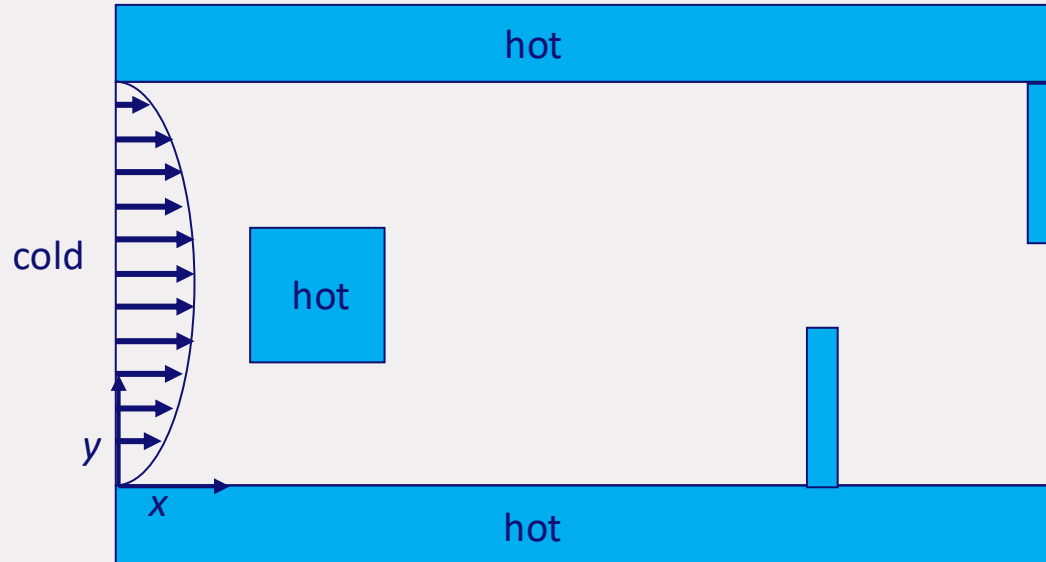
Introduction to CFD (4RC30)

Boundary conditions (section 2.10 & 2.11, Chapter 9)

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What/where are the BC's



- inlet
- outlet
- wall
- internal wall

BC for every
solving variables!

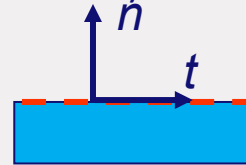
BC's: overview

Commonly used BC	Feature
Dirichlet Boundary Condition	Fixed value of the solution (e.g. $T = A$) <i>No-slip BC is a Dirichlet application</i>
Neumann Boundary Condition	Fixed derivative of the solution (e.g. $\partial u / \partial n = B$) <i>Free-slip BC is a Neumann application</i>
Mixed Boundary Condition	Combined use of Dirichlet and Neumann BC <i>(example next slide)</i>
Robin Boundary Condition	Setting a linear combination of both the value and its derivative (e.g. heat loss through a surface (derivative) depends directly on the surface's actual temperature (value))

- **Initial conditions** ($t = 0$): special BC
Everywhere in solution region given $\rho, \mathbf{u}, T, k, \varepsilon, \dots$

BC's: overview

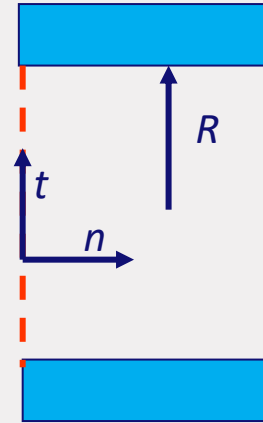
- **BC's on solid walls:**
 - $\mathbf{u}_t = \mathbf{u}_w$ (no slip condition, mostly $\mathbf{u}_w = 0$)
 - $T = T_w$ (fixed wall temperature)
 - $k \partial T / \partial n = -q_w$ (fixed heat flux)
- **BC's on free surfaces:**
 - $u_n = 0$ (no fluid cross the boundary)
 - $\partial u_t / \partial n = 0$ (frictionless “wall”, free slip condition)



BC's at inlet

- Given profiles of $u, T, k, \varepsilon, \dots$
 - **Example:**
 - Constant profile for T_{in}
 - Parabolic or plug flow profile for $u_{in,n}$
 - Straight inflow: $u_{in,t} = 0$
- $$k_{in} = \frac{3}{2}(u_{in}T_i)^2; \varepsilon_{in} = C_\mu^{3/4} \frac{k^{3/2}}{l}; l = 0.07R$$

T_i is turbulence intensity (1-6%)



BC's at outlet

- **Stress continuity:**

$$-p + \frac{\mu \partial u_n}{\partial n} = F_n \qquad \frac{\mu \partial u_t}{\partial n} = F_t$$

- **In fully developed flow:**

$$\frac{\partial u_n}{\partial n} = \frac{\partial u_t}{\partial n} = 0 \Rightarrow F_n = -p \qquad F_t = 0$$

- **Most used: outflow pressure BC (flow finish developing)**

$$p = p_{outlet} \qquad \frac{\partial u_n}{\partial n} = 0 \qquad \frac{\partial T}{\partial n} = 0 \qquad \frac{\partial k}{\partial n} = 0 \qquad \frac{\partial \varepsilon}{\partial n} = 0$$

BC's at outlet

- In real-world simulations, outlet often suffer from **reverse flow (backflow)**, a strict static condition can easily crash the solver.
- Sometimes not a prior known if **flow in or out** (e.g. windows in a room)

Use a dynamic BC: e.g. **inletOutlet** Boundary Condition (in OpenFOAM)

- reads the direction of the local fluid flux (Φ) at grids face the boundary
- **If $\Phi > 0$, fluid leaving the domain (outflow): Neumann/zero-gradient condition**
- **If $\Phi < 0$, fluid entering the domain (inflow/backflow): Dirichlet/fixedValue condition using a user-defined inletValue**

Overall mass balance

The overall mass balance can be satisfied by:

$$M_{in} = \sum_{j=J_{in,min}}^{J_{in,max}} A_{cell} \rho u$$

$$M_{out} = \sum_{j=J_{out,min}}^{J_{out,max}} A_{cell} \rho u$$

$$u_{NPI+1,J} = u_{NPI,J} \cdot \frac{M_{in}}{M_{out}}$$

[bound.m](#)

```
for J = 2:NPI+1
```

```
    j = J;
```

```
    AREAw = y_v(j+1) - y_v(j);
```

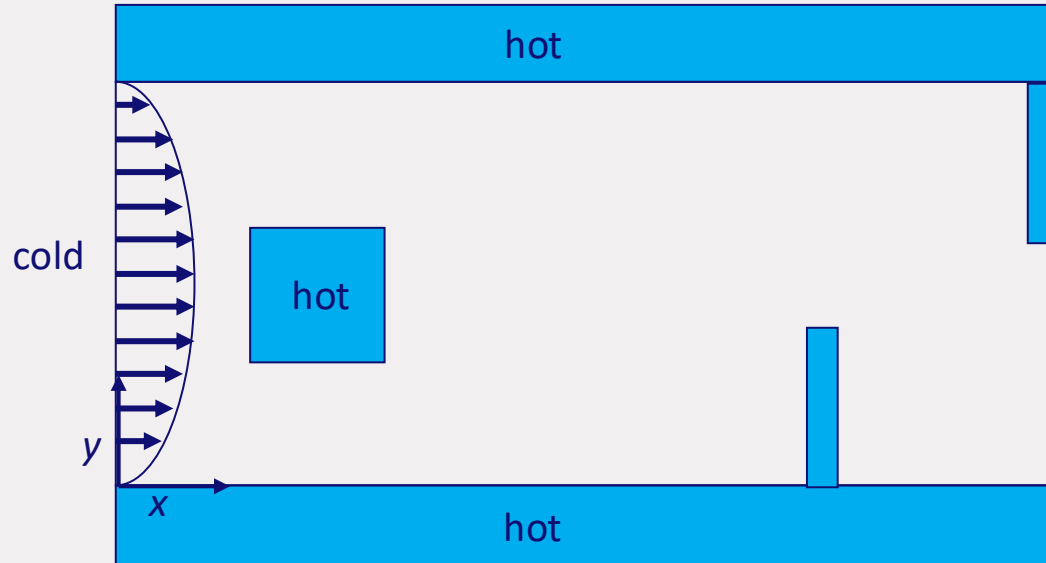
```
    m_in   = m_in  + F_u(2,J)*AREAw;
```

```
    m_out  = m_out + F_u(NPI+1,J)*AREAw;
```

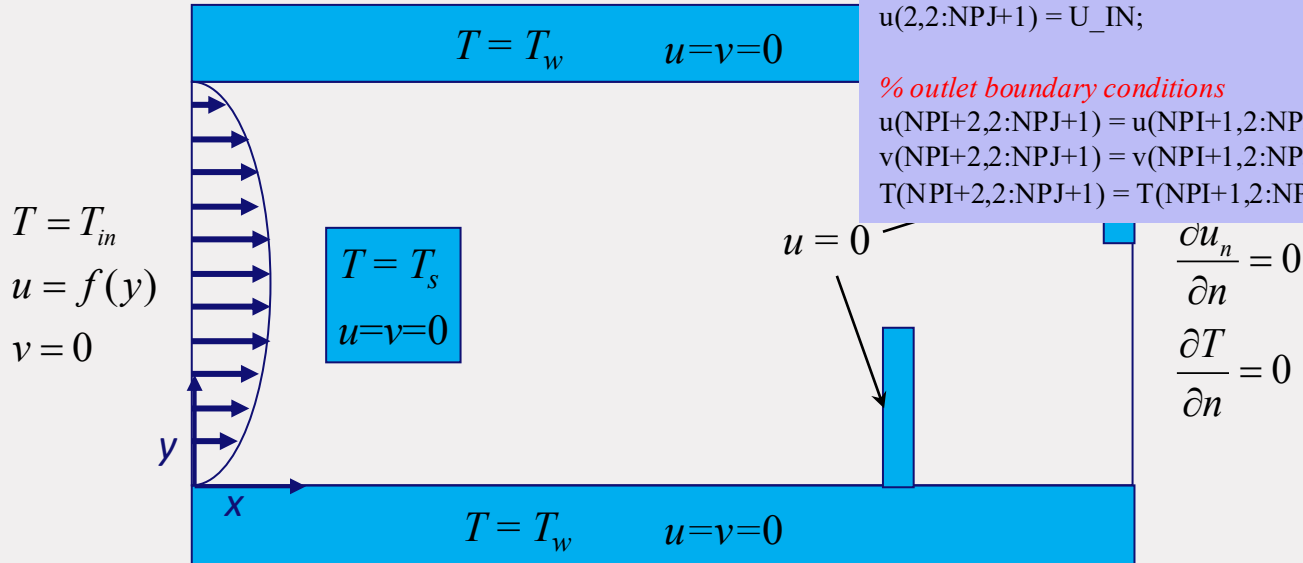
```
end
```

```
u(NPI+2,2:NPI+1) = u(NPI+1,2:NPI+1)*m_in/m_out;
```


What are the BC's



What are the BC's



bound.m

% Fixed temperature at the upper and lower wall

$T(1:NPI+2,1) = 373.;$ *% lower wall*

$T(1:NPI+2,NPJ+2) = 373.;$ *% upper wall*

% Fixed temperature/velocity of the incoming fluid

$T(1,2:NPJ+1) = 273.;$

$u(2,2:NPJ+1) = U_IN;$

% outlet boundary conditions

$u(NPI+2,2:NPJ+1) = u(NPI+1,2:NPJ+1)*m_in/m_out;$

$v(NPI+2,2:NPJ+1) = v(NPI+1,2:NPJ+1);$

$T(NPI+2,2:NPJ+1) = T(NPI+1,2:NPJ+1);$

BC's internal walls

- Internal walls are inside solution domain, all the equations are solved
- Use source terms to set variables to fixed value:

$$S_P = -10^{30} \quad S_u = 10^{30} \phi_{fix}$$

$$(a_P - S_P) \phi_P = \sum a_{nb} \phi_{nb} + S_u$$

$$(\cancel{a_P} + 10^{30}) \phi_P = \sum \cancel{a_{nb}} \phi_{nb} + 10^{30} \phi_{fix}$$

$$\phi_P = \phi_{fix}$$

ucoeff.m

```
LARGE = 1E30;
```

```
if i == NPI/5 && J < NPJ/3
```

```
    SP[i][J] = -LARGE;
```

```
    Su[i][J] = LARGE*u_fix;
```

```
end
```

Boundary conditions $k - \varepsilon$ at walls

- On solid walls high shear stress
- Equilibrium between k and ε

- Friction or shear velocity $u_\tau = \left(\frac{\tau_w}{\rho} \right)^{1/2}$

- Dimensionless wall distance $y^+ = \frac{\rho u_\tau y}{\mu}$

- Dimensionless velocity $u^+ = \frac{U}{u_\tau}$

Law of the wall

Boundary layer exists of:

- Viscous sub-layer (**thin!**) 

Physical viscosity of the fluid kills the turbulence

$$y^+ < 11.63; \quad u^+ = y^+$$

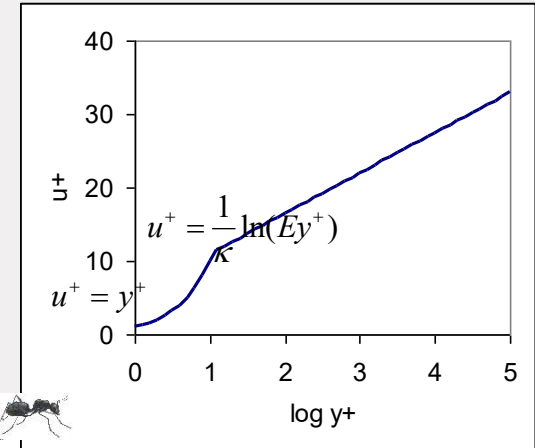
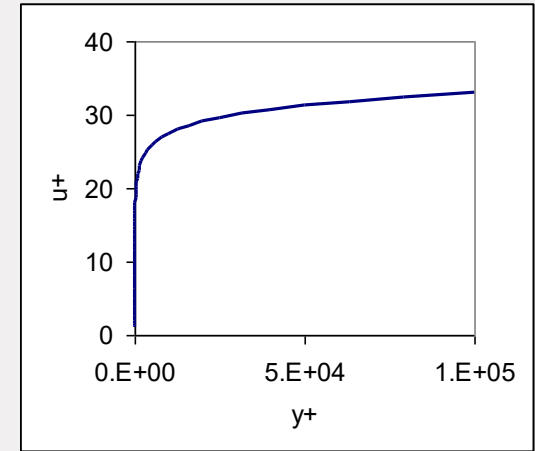
- Turbulent layer: log-law region

$$y^+ > 11.63; \quad u^+ = \frac{1}{\kappa} \ln(Ey^+)$$

Von Kármán constant $\kappa = 0.4184$,

Wall roughness constant $E = 9.793$

Avoiding extreme fine mesh down to the wall, **wall functions** are used with the first grid cell placed in the turbulent layer.



Implementation law of the wall (1)

Velocity (U) Wall Function

- u velocity parallel to wall (governed by the log-law):

$$u^+ = \frac{1}{\kappa} \ln(Ey^+)$$

$$\tau_w = \frac{\rho C_\mu^{1/4} k^{1/2} u_p}{u^+}$$

- Implementation as source terms:

$$a_S = 0; \quad S_{P,u} = -\frac{\rho C_\mu^{1/4} k^{1/2}}{u^+} A_{Cell}$$

At the wall, no-slip or zero velocity.

Implementation law of the wall (2)

Turbulent Kinetic Energy (k) Wall Function (cells adjacent to the wall)

- k is calculated dynamically, but its production and dissipation are assumed to be in a state of local equilibrium.
- The friction velocity u_τ is linked to k :

$$u_\tau = C_\mu^{1/4} \sqrt{k}$$

$$P_k^{wall} = \frac{\rho u_\tau^3}{\kappa y}$$

- Implementation as source terms:

$$a_S = 0; \quad S_{P,k} = -\frac{\rho C_\mu^{3/4} k_{old}^{1/2} u^+}{\Delta y_P} \Delta V; \quad S_{u,k} = \frac{\tau_w u_P}{\Delta y_P} \Delta V$$

At the wall, apply a **zero-gradient** (Neumann) condition: $\partial k / \partial n = 0$

Implementation law of the wall (3)

Turbulence Dissipation Rate (ε) Wall Function (cells adjacent to the wall)

- ε is explicitly overwritten
- fixed with a Dirichlet value calculated directly from the cell's local kinetic energy :

$$\varepsilon = \frac{C_\mu^{3/4} k^{3/2}}{\kappa y}$$

- Implementation as source terms:

$$S_{P,\varepsilon} = -10^{30}; \quad S_{u,\varepsilon} = \frac{C_\mu^{3/4} k_P^{3/2}}{\kappa \Delta y_P} \cdot 10^{30}$$

Implementation law of the wall

- u velocity parallel to wall:

$$a_S = 0; \quad S_{P,u} = -\frac{\rho C_\mu^{1/4} k^{1/2}}{u^+} A_{Cell}$$

- k equation:

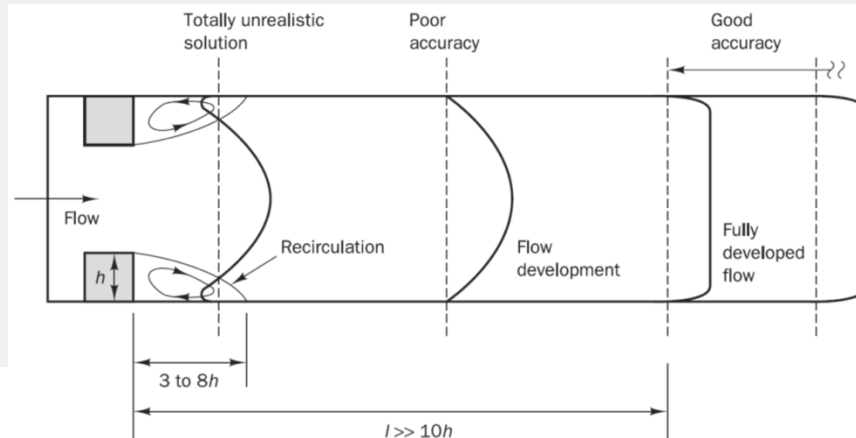
$$a_S = 0; \quad S_{P,k} = -\frac{\rho C_\mu^{3/4} k_{old}^{1/2} u^+}{\Delta y_P} \Delta V; \quad S_{u,k} = \frac{\tau_w u_P}{\Delta y_P} \Delta V$$

- ε equation:

$$S_{P,\varepsilon} = -10^{30}; \quad S_{u,\varepsilon} = \frac{C_\mu^{3/4} k_P^{3/2}}{\kappa \Delta y_P} \cdot 10^{30}$$

What does NOT work?

- One inlet, but no outlets
- No developed/circulating flow at the outlet (unless a dynamic BC used)



What does NOT work?

Over-constrained BCs

- **Example 1:**

An incompressible fluid flows through a pipe with constant diameter:

- Inlet: Dirichlet BC. $u = 5$ m/s
- Outlet: Dirichlet BC. $u = 3$ m/s

Mass conservation failed

- **Example 2: The Pressure-Velocity Conflict**

Air flows through a high-speed converging nozzle:

- Inlet: Dirichlet BC. $u = 50$ m/s, $p = 101$ Pa
- Outlet: open BC

Velocity and pressure are linked by momentum conservation. You can specify the velocity *or* you can specify the pressure, but **you cannot dictate both**.